

Bari Evidence Registry Memos

Bari Evidence Registry Memo: E160b, Annatto (Bixin / Norbixin)

Executive Conclusion

Annatto (E160b) is a naturally derived pigment extracted from the outer seed coat of the *Bixa orellana* tree, historically utilized to confer yellow to orange-red hues to dairy products, baked goods, and fats. The regulatory landscape surrounding annatto is defined by the toxicokinetic divergence of its two primary molecules: bixin, which is fat-soluble, and norbixin, which is water-soluble. The European Food Safety Authority (EFSA) completed comprehensive toxicological re-evaluations in 2016 and 2019, concluding that while neither compound is genotoxic or carcinogenic, norbixin is absorbed at significantly higher rates than bixin, thereby necessitating distinct safety thresholds¹. Consequently, EFSA established an Acceptable Daily Intake (ADI) of 6 mg/kg body weight (bw) per day for bixin and a highly restrictive ADI of 0.3 mg/kg bw per day for norbixin¹. This assessment revealed a critical over-exposure concern specifically regarding norbixin at the 95th percentile of consumption among toddlers and children, prompting the European Commission to enact Regulation (EU) 2020/771, which formally bifurcated the additive into Annatto bixin (E160b(i)) and Annatto norbixin (E160b(ii)) and reduced maximum permitted levels in key dietary staples³. In Israel, the Ministry of Health (MoH) permits E160b across various matrices, capping usage at 10 mg/kg in margarines, 15 mg/kg in processed and ripened cheeses, and 25 ppm in dietary supplements⁵. The bottom-line evidence verdict positions annatto as fundamentally safe for the general population under current statutory limits. However, clinical literature confirms a valid, albeit statistically rare, Immunoglobulin E (IgE)-mediated hypersensitivity signal. Because this allergy risk represents an idiosyncratic vulnerability rather than a population-level toxicity, the additive does not warrant a severe penalty but strictly requires contextual allergen disclosure.

Evidence Table

Source	Org/Journal	Year	URL/DOI /PMID/Doc No.	Key points	Evidence strength	Registry implication
Scientific Opinion on the safety of	EFSA Journal	2016	DOI: 10.2903/j.efsa.2016	Evaluated subchronic toxicity,	High (Regulatory Consensus)	Establishes the foundational safety

annatto extracts (E 160b)			.4544	genotoxicity, and carcinogenicity. Derived separate ADIs for bixin (6 mg/kg bw) and norbixin (0.3 mg/kg bw). Identified that dietary exposure for norbixin exceeded the ADI at the 95th percentile for toddlers.	us)	parameters and dictates the toxicological distinction between the fat-soluble and water-soluble extracts.
Commission Regulation (EU) 2020/771	European Commission	2020	CELEX:32020R0771	Legally separated E160b into E160b(i) bixin and E160b(ii) norbixin. Lowered maximum permitted	High (Statutory Law)	Highlights the regulatory agility required to prevent cumulative over-exposure, anchorin

				d levels (MPLs) in specific food categories to mitigate the pediatric over-exposure to norbixin identified in the 2016 EFSA evaluation.		g the additive's safety to strict dosage compliance.
Annatto hypersensitivity after oral ingestion	Ann Allergy Asthma Immunol	2021	PMID: 34320336	Documented isolated cases of IgE-mediated hypersensitivity and urticaria to annatto extracts. Confirmed the pathogenesis via rigorous placebo-	Moderate (Clinical Case Series)	Validates the clinical reality of an individual-level allergy risk, justifying an allergy caution without penalizing the additive's population-level safety

				controlled oral challenges at realistic dietary doses.		score.
Allergy for cheese: evidence for an IgE-mediated reaction from annatto	Allergy	2009	PMID: 19772519	Demonstrated basophil activation and IgE-mediated anaphylaxis tracing back directly to the natural dye annatto utilized in cheese matrices, distinguishing it from dairy protein allergies.	Moderate (Clinical Case Series)	Provides the immunological mechanism for unexplained adverse reactions to dairy products, proving annatto acts as a functional hapten.

Risk Interpretation

The physiological and toxicological profile of annatto requires a bifurcated analysis separating general population toxicity from specific immunological hypersensitivity. From a foundational safety perspective, the active carotenoid compounds in annatto—bixin and norbixin—exhibit a unique linear, non-cyclic chemical structure that distinguishes them from provitamin A

carotenoids¹. EFSA's extensive subchronic in vivo studies demonstrated no treatment-related mortality, genotoxic damage, or carcinogenic initiation at standard doses². The critical regulatory pivot occurred when toxicokinetic data revealed that norbixin is absorbed and systemically distributed at far higher efficiencies than bixin. This disparity forced EFSA to assign norbixin an ADI twenty times more restrictive than bixin (0.3 mg/kg bw versus 6.0 mg/kg bw)¹. Dietary exposure modeling based on these thresholds indicated that heavy consumers of processed foods—particularly toddlers at the 95th percentile of intake—were breaching the norbixin ADI. The European Commission mitigated this specific over-exposure concern by drafting Regulation (EU) 2020/771, which structurally divided the additive and depressed the legal maximum permitted levels across the food supply³. Within the Israeli market, the Ministry of Health mirrors these tight tolerances, strictly capping E160b concentrations in high-consumption vectors like cheeses and margarines⁵. Consequently, provided manufacturers adhere to these statutory limits, the risk of chronic toxicological over-exposure is adequately neutralized.

Conversely, the immunological risk profile of annatto requires distinct interpretation. While the vast majority of consumers process the pigment without incident, clinical allergology confirms that annatto can act as a potent hapten. Because the molecule is relatively small (395 daltons), it must bind to carrier proteins—such as bovine serum albumin or kappa-casein in cheese matrices—to crosslink IgE antibodies and precipitate an immune response⁹. This mechanism drives true Type I hypersensitivity, distinct from non-IgE food intolerances or IgG pathways. Documented clinical manifestations include acute urticaria, angioedema, and, in highly sensitized individuals, anaphylaxis¹⁰. Placebo-controlled oral challenges and flow cytometric basophil activation tests have irrefutably validated this pathway at realistic dietary exposures¹². However, epidemiological tracking places the prevalence of this allergy at an exceptionally low rate. Similar to the hypersensitivity profile of carmine (E120), annatto allergy represents a rare, idiosyncratic immunological error rather than evidence of inherent toxicity¹⁴. Therefore, the population-level safety rating should not be degraded, provided the ingredient is clearly disclosed for sensitive individuals to avoid.

Regarding dietary preferences, annatto is inherently plant-derived, extracted solely from the seeds of the *Bixa orellana* shrub. This origin ensures complete compatibility with vegan and vegetarian diets. Furthermore, absent cross-contamination with non-compliant solvents or carriers during the extraction process, annatto inherently satisfies the requirements for Kosher and Halal certification¹⁵.

To maintain scientific integrity and prevent consumer alarmism, several exaggerated claims must be explicitly avoided. It is inaccurate to describe annatto as a "toxic" or "dangerous" chemical, as regulatory bodies globally recognize it as a safe natural colorant when deployed within ADI limits. Furthermore, annatto must not be conflated with synthetic azo dyes, such as Tartrazine (E102) or Allura Red (E129); annatto lacks the neurodevelopmental signals associated with those synthetics and is entirely exempt from the European Union's hyperactivity warning label requirements for pediatric populations¹⁵. Finally, while the allergenic

potential is real, it is statistically incorrect to label annatto a "common" food allergen, as its prevalence is dwarfed by the major recognized allergen groups.

Bari Tier Recommendation

Assigned Tier: likely-neutral **Justification:** Annatto possesses a robust history of safe application as a natural colorant, and while an isolated, rare IgE-mediated hypersensitivity signal exists alongside strict ADI constraints for its norbixin fraction, it lacks the broad, mechanistic harm pathways required to trigger a more severe penalty. **Adjustment Criteria:** This tier could be downgraded to dose-dependent if future market-basket analyses within the Israeli retail sector reveal systemic manufacturer breaches of the norbixin ADI in ultra-processed pediatric foods. Conversely, it could be upgraded to functional if ongoing immunological research conclusively demonstrates that the haptenization process requires volumetric exposures magnitudes above standard dietary consumption, thereby entirely neutralizing the real-world allergy risk.

Registry-Ready Entry

YAML

```
signal_id: additive_E160b_annatto_color
e_number: E160b
canonical: annatto_bixin_norbixin
ingredient_names:
  - Annatto
  - Bixin
  - Norbixin
  - Annatto extracts
  - CI Natural Orange 4
  - INS 160b
  - E160b(i)
  - E160b(ii)
hebrew_patterns:
  "אָנאָטוֹ|אָנאָטוֹת מַצִּית+|טִבְעִי+|מֵאֵכֶל+|צִבְעוֹן|E[\s-]*160[\s-]*[bB]|אָנאָטוֹ|בִּיקְסִין|נורבִּיקְסִין"
english_patterns: "(?i)annatto|bixin|norbixin|e[\s-]?160b|natural\s+orange\s+4"
additive_class: colorant_natural
proposed_d4_tier: likely-neutral
evidence_level: high
scoring_effect:
  nutrition_score: no_change
  additive_complexity: mild_flag
```


baked commodities, commercial juices, and alcoholic beverages to arrest enzymatic browning and inhibit bacterial proliferation. The toxicological consensus regarding sulphites experienced a seismic shift during the 2022 EFSA re-evaluation. Citing unresolved gaps in neurotoxicity and developmental toxicity data, EFSA completely withdrew the temporary group Acceptable Daily Intake (ADI) and implemented a Margin of Exposure (MOE) risk assessment methodology¹⁶. The resulting models demonstrated that typical dietary exposure significantly breaches the safe MOE threshold across nearly all demographic populations, indicating a tangible safety concern at realistic food-use doses¹⁸. Concurrently, sulphites operate as a universally declared allergen; exposures exceeding 10 mg/kg (or 10 mg/L) of SO₂ reliably precipitate severe bronchospasm in susceptible asthmatic populations¹⁹. In Israel, the Ministry of Health maintains category-specific maximums—such as 200 mg/L in wines and up to 340 ppm in dried ginger—while the domestic SI 1145 standard strictly enforces the 10 mg/kg allergen labeling mandate mirroring European directives⁵. Given the intersection of a confirmed acute asthmatic harm pathway and a systemic regulatory MOE breach at non-extreme dietary exposures, the sulphite family is assigned a heavily penalized verdict.

Evidence Table

Source	Org/Journal	Year	URL/DOI /PMID/Doc No.	Key points	Evidence strength	Registry implication
Follow-up of the re-evaluation of sulfur dioxide (E 220)...	EFSA Journal	2022	DOI: 10.2903/j.efsa.2022.7594	Withdrew the historical ADI. Established a Benchmark Dose Lower Confidence Limit (BMDL) of 38 mg SO ₂ eq/kg bw/day. Concluded that the	High (Regulatory Consensus)	Provides the definitive toxicological basis for treating typical food-dose exposures of sulphites as a measurable population-level

				Margin of Exposure (MOE) fell below the safe threshold of 80 for nearly all populations at standard dietary intakes.		health concern.
Joint EFSA–ECHA document	EFSA/ECHA	2022	EFSA–ECHA–2022–SO2	Reconciled biocide versus food additive assessments. Confirmed that dietary exposure to sulphite texturizers breaches safety margins based specifically on neurotoxic endpoints	High (Inter-agency Report)	Validates the severity of the systemic toxicological signal distinct from the acute asthma pathway, solidifying the MOE penalty.

				(delayed visual evoked potentials).		
Draft Public Health Regulations (Food) (Food Labeling)	Israel MoH	2021	MoH Reg Draft 2021	Adopts European SI 1145 standards, requiring explicit, bolded allergen declarations within the ingredient list for any sulphite concentrations exceeding 10 ppm (mg/kg or mg/L).	High (Statutory Law)	Dictates the absolute legal necessity to uniformly flag E220–E228 as declared allergens within the Israeli retail market architecture.

Risk Interpretation

The risk interpretation for the sulphite family (E220–E228) mandates a dual-pronged analysis, addressing both chronic systemic toxicity for the general population and acute hypersensitivity for vulnerable subgroups. Historically, sulphites were managed under a temporary group ADI of 0.7 mg/kg bw/day. However, the EFSA 2022 re-evaluation fundamentally destabilized this framework. Because critical literature gaps regarding biological and neurotoxicological endpoints could not be resolved by industry stakeholders, EFSA formally withdrew the ADI entirely¹⁶. In its place, the authority utilized a Margin of Exposure (MOE) approach. Toxicologists derived a new reference point—a Benchmark Dose Lower

Confidence Limit (BMDL) of 38 mg SO₂ equivalents/kg bw/day—based on clinical observations of prolonged visual evoked potential latency, a distinct indicator of neurotoxicity¹⁸. When applying the requisite assessment factor of 80, the modeled MOE must exceed 80 to be considered safe. Alarming, refined exposure scenarios analyzing actual food consumption patterns revealed that the MOE was consistently breached (<80) across toddlers, children, and adults¹⁷. Within the Israeli dietary context, the cumulative consumption of dried fruits (which legally contain hundreds of ppm of SO₂ to preserve color), commercial fruit juices, and wine guarantees that this MOE is routinely compromised at mean, non-extreme exposure levels⁵. Separate from the chronic MOE concern is the acute, localized harm pathway characterized by sulphite-induced asthma. Sulphites are universally recognized as a declared allergen. While true IgE-mediated anaphylaxis to sulphites is extraordinarily rare, non-IgE hypersensitivity mechanisms impact up to 10% of the global asthmatic population²². The physiological pathogenesis occurs during mastication; the warm, acidic environment of the human oral cavity rapidly converts sulphite salts into sulphur dioxide gas. The inhalation of this localized gas triggers immediate and severe bronchospasm in hyper-reactive airways. Because this mechanism constitutes a confirmed harm pathway at highly relevant food doses, both global regulatory bodies and Israeli statutory law (SI 1145) enforce uncompromising labeling thresholds, mandating bolded allergen warnings for any concentration equal to or exceeding 10 mg/kg¹⁹.

Regarding dietary preferences, the sulphite family is produced via synthetic chemical processes or mineral derivation. The molecules themselves do not inherently violate Vegan, Vegetarian, Kosher, or Halal doctrines. However, the presence of sulphur dioxide (E220) is ubiquitous in viniculture, and its application in wine matrices frequently intersects with stringent regulations governing specific Kosher wine certifications.

To maintain precision in consumer communications, unsupported and exaggerated claims must be filtered. It is medically inaccurate to claim that sulphites cause widespread anaphylaxis in the general population; the primary acute vector is strictly asthmatic bronchospasm. Furthermore, the chemical family must not be conflated with "sulfates," which are generally inert nutritional compounds commonly found in water supplies and supplements. Finally, one must avoid the pervasive myth that organic or "natural" wines are universally sulphite-free. The fermentation of yeast naturally generates trace amounts of endogenous sulphites (often yielding 10 to 20 ppm), meaning absolute zero-sulphite claims are biologically improbable and often misleading.

Bari Tier Recommendation

Assigned Tier: confirmed-negative **Justification:** The sulphite family meets the highest evidentiary threshold for penalty. The harm pathway for asthmatic bronchospasm is a confirmed, declared reality at relevant food-use exposures. Crucially, the 2022 EFSA withdrawal of the ADI and the calculation of a neurotoxicity-based Margin of Exposure (MOE) breach at *typical* mean dietary consumption elevates the systemic hazard from contested to a credible weight-of-evidence harm. **Adjustment Criteria:** If regulatory bodies enact sweeping

legislative phase-outs or drastically reduce the maximum permitted levels to a point where the MOE is no longer breached in typical diets, the general-population tier could theoretically be downgraded to dose-dependent. Conversely, the allergen status remains immutable barring a complete eradication of the chemical from the food supply.

Registry-Ready Entry

YAML

signal_id: additive_E220_sulphite_family

```
e_number: E220
```

canonical: sulphur_dioxide_sulphites

ingredient_names:

- Sulphur dioxide

- Sodium sulphite

- Sodium bisulphite

- Sodium metabisulphite

- Potassium metabisulphite

- Calcium sulphite

- Calcium bisulphite

- Potassium bisulphite

- INS 220

- INS 221

- INS 222

– INS 223

- INS 224

- INS 225

- INS 228

hebrew patterns:

2[0-8]"

english_patterns:

```
"(?:sulphur\s+dioxide|sulfur\s+dioxide|sulphite|sulfite|bisulphite|bisulfite|metabisulphite|metabisulfite|e[\s]?22[0-8])"
```

additive class: preservative sulphite

proposed d4 tier: confirmed-negative

evidence level: high

scoring effect:

nutrition score: no change

additive complexity: strong flag

surfactants predominantly utilized to stabilize emulsions, manipulate viscosity, and inhibit crystallization in baked goods, ice creams, and industrial fat matrices. From a classical toxicological perspective, global regulatory bodies view these agents as fundamentally benign. EFSA re-evaluations have confirmed their legality at standard dietary doses, establishing an ADI of 10 mg/kg bw for polysorbates (E435), concluding no numerical ADI was necessary for E475 due to its breakdown into absorbable lipids, and assigning 25 mg/kg bw for E477²⁴. Concurrently, the Israel Ministry of Health (MoH) permits aggressive utilization of these compounds, authorizing levels up to 5–10 g/kg in pastries, emulsified sauces, and granola⁵. However, modern epidemiological and mechanistic paradigms severely destabilize this classical consensus. Rigorous prospective cohort data from the NutriNet-Santé project (2023–2024) specifically correlates the intake of polyglycerol esters (E475) with statistically significant hazard ratios for Type 2 Diabetes²⁹. Mechanistically, Polysorbate 60 (E435) exhibits structural homology to Polysorbate 80 (E433), operating as a potent detergent that ablates the mucosal gut barrier, inciting microbiome dysbiosis and low-grade systemic inflammation³¹. Consequently, the deeply contested evidence surrounding ultra-processed texturizers firmly extends to these individual molecular agents.

Evidence Table

Source	Org/Journal	Year	URL/DOI /PMID/Doc No.	Key points	Evidence strength	Registry implication
Food emulsifiers and risk of type 2 diabetes (NutriNet-Santé)	Lancet Diabetes Endocrinol	2024	DOI: 10.1016/S2213-8587(24)00086-X	Epidemiological cohort analysis identified significant positive hazard ratios linking Type 2 Diabetes directly to the consumption of "total	High (Prospective Cohort)	Quantitatively anchors E475 to metabolic dysregulation and insulin resistance at population-level dietary exposures.

				polyglycerol esters of FAs" (E475).		
Re-evaluation of polyglycerol esters of fatty acids (E 475)	EFSA Journal	2017	DOI: 10.2903/j.efsa.2017.5089	Classical toxicological evaluation found no acute toxicity, genotoxicity, or carcinogenicity in subchronic murine models. Concluded no numerical ADI was required.	High (Regulatory Consensus)	Documents the conflicting baseline regulatory stance that permits widespread industrial usage, establishing the "contested" nature of the additive.
Food additives: Assessing the impact of exposure. ..	Aliment Pharmacol Ther	2019	PMID: 31631437	Identifies the broader polysorbate family (including PS60/E435) as detergent-like agents capable of disrupting	Moderate (Mechanistic / In Vivo)	Biologically connects E435 to the established mucosal-disruption pathogenesis famously associated

				g the intestinal mucus layer, driving bacterial translocation and metabolic syndrome.		d with Polysorbate 80 (E433).
Food additive emulsifiers and cancer risk (NutriNet-Santé)	PLoS Med	2024	DOI: 10.1371/journal.pmed.1004338	While primarily highlighting E471 and carrageenans, the cohort analysis groups synthetic emulsifiers as a cocktail of texturizers linked broadly to chronic inflammation and extraintestinal malignancies.	High (Prospective Cohort)	Confirms the long-term, low-grade inflammatory hazard of complex synthetic emulsifier cocktails deployed in UPF architectures.

Risk Interpretation

The toxicological assessment of E435, E475, and E477 embodies the current fracture between classical food safety frameworks and modern microbiome science. Regulatory bodies historically classified these emulsifiers based exclusively on acute systemic toxicity assays. For example, EFSA noted that E475 is readily hydrolyzed in the gastrointestinal tract into polyglycerols and free fatty acids, which the body metabolizes similarly to natural dietary fats, thereby negating the need for a numerical ADI²⁵. Similarly, E477 is hydrolyzed into fatty acids and propane-1,2-diol (propylene glycol), which clears standard toxicity thresholds to earn a 25 mg/kg bw ADI²⁶. E435, a polyoxyethylene sorbitan ester, shares a 10 mg/kg bw group ADI with other polysorbates²⁷. Because these molecules do not induce acute organ failure, mutagenesis, or tumor initiation in classic 90-day rat studies, the prevailing regulatory conclusion is absolute safety at typical food doses.

However, the vanguard of epidemiological and microbiological research fundamentally challenges this assumption, demonstrating that the primary harm vectors for these compounds are not systemic poisoning, but localized biophysical disruption within the gastrointestinal tract. The evidence against the broader emulsifier class absolutely extends to these individual agents through both direct epidemiological hazard ratios and structural mechanistic homology.

- **E475 (Polyglycerol Esters):** In the landmark NutriNet-Santé cohort analysis published in the *Lancet Diabetes & Endocrinology* (Salame et al., 2024), researchers utilized massive dietary datasets cross-referenced against brand-specific additive matrices. They discovered that consumption of "total polyglycerol esters of FAs," which prominently features E475, yielded an increased hazard ratio (HR = 1.34 [1.10-1.63], $p < 0.001$) for incident Type 2 Diabetes²⁹. This represents a direct, population-level correlation between this specific emulsifier and metabolic disease.
- **E435 (Polysorbate 60):** The mechanistic disruption of the gut-barrier established by Chassaing et al. for Polysorbate 80 (E433) is fundamentally translatable to Polysorbate 60. Both compounds are polyoxyethylene sorbitan esters acting as highly potent non-ionic surfactants. In vivo models repeatedly demonstrate that these detergents aggressively thin the protective mucosal layer lining the colon. This physical degradation permits bacterial flagellin to penetrate the epithelium, triggering a chronic, low-grade inflammatory response that serves as the biological precursor to obesity and metabolic syndrome³¹.
- **E477 (PGMS):** While less frequently isolated in discrete cohort hazard ratios compared to E471 or E475, E477 operates within the exact same texturizing matrix of ultra-processed foods. Given the NutriNet-Santé findings that industrial emulsifier *mixtures* (cocktails) heavily exacerbate cardiometabolic risks via synergistic dysbiosis, E477 inherently inherits the class-wide concern regarding mucosal inflammation and barrier dysfunction³⁵.

Regarding dietary preferences, the manufacturing profiles of E435, E475, and E477 generate significant ambiguity. These emulsifiers are synthesized via the esterification of fatty acids (such as stearic, palmitic, and oleic acids). These source fatty acids can be derived from plant oils (e.g., soybean or palm oil) or rendered from animal fats (e.g., bovine tallow)²⁷. Because standard ingredient labels simply read "Polyglycerol esters" or "Polysorbate 60," it is structurally impossible for a consumer to ascertain the biological origin without a verified Vegan, Halal, or Kosher certification from the specific manufacturer.

To preserve scientific rigor, unsupported or exaggerated claims must be excised from consumer communications. It is medically inaccurate to state that these specific emulsifiers definitively *cause* cancer; the NutriNet-Santé cohort primarily established oncological links for E471 and carrageenans, whereas E475 is predominantly linked to Type 2 Diabetes³⁰.

Furthermore, it is alarmist to describe them as "toxic chemicals." Their harm profile is derived from the physical disruption of gut biome mechanics (surfactant effects) and subsequent metabolic cascades, not acute systemic poisoning.

Bari Tier Recommendation

Assigned Tier: contested **Justification:** There is a highly credible, published mechanistic and clinical signal linking these specific texturizing emulsifiers (especially E435 via mucosal degradation and E475 via explicit NutriNet-Santé hazard ratios) to gut-barrier disruption and metabolic syndrome. However, classical regulatory agencies (EFSA, FDA, Israel MoH) continue to maintain their safety status, meaning the emerging scientific consensus directly contests the statutory reality. **Adjustment Criteria:** This tier would escalate to confirmed-negative if global regulatory bodies formally adopt microbiome-disruption endpoints into their risk assessment frameworks and consequently revise the ADIs downward or institute targeted bans. The tier would drop to likely-neutral only if long-term randomized controlled clinical trials definitively exonerate these specific molecules from inducing mucosal damage in humans.

Registry-Ready Entry

YAML

signal_id: additive_emulsifier_trio_E435_E475_E477

e_number: "E435, E475, E477"

canonical: synthetic_emulsifiers_polysorbate_pge_pgms

ingredient_names:

- Polysorbate 60

- Polyoxyethylene sorbitan monostearate

- Polyglycerol esters of fatty acids

emulsification. However, the additive complexity index must be elevated significantly. These synthetic surfactants are powerful industrial texturizers that definitively flag a product as an Ultra-Processed Food (UPF). A strict dietary preference caution must be triggered, warning Vegan, Kosher, and Halal consumers that the constituent fatty acids possess an ambiguous biological origin (animal vs. plant) unless the product carries an explicit overriding certification. No allergy caution is required, as these lipids do not trigger IgE mechanisms.

Consumer Wording:*English:* "Contains synthetic emulsifiers (E435, E475, or E477). While legally permitted by food safety authorities, recent large-scale research has linked heavy, long-term consumption of these industrial texturizers to negative changes in gut bacteria, inflammation, and an increased risk of metabolic issues such as Type 2 Diabetes." *Hebrew:* למרות שהם מאושרים לשימוש על ידי רשויות (E435, E475 או E477). מכיל מתחלבים תעשייתיים הבריאות, מחקרים רחבי-היקף מהעת האחרונה מקשרים צריכה מוגברת ומושכת של חומרים אלו 2. לפגיעה בחיידקי המעי, לדלקתיות, ולסיכון מוגבר לבעיות מטבוליות כגון סוכרת סוג

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